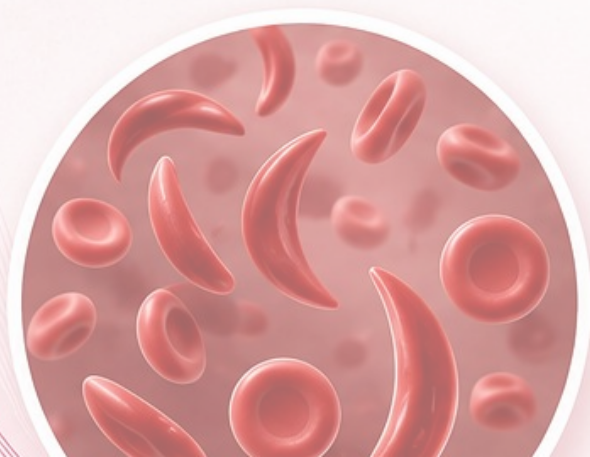




SICKLE CELL *in* PREGNANCY



High Risk Pregnancy



Prevent Complications



Ensure Maternal &
Fetal Well-being



Topic 5



TYPES OF SICKLE CELL DISEASE



HbSS (SICKLE CELL DISEASE)

People who have HbSS inherit sickle cell genes (S) from both parents and have sickle cell disease (SCD). They have various complications and requires lifelong management.



SICKLE CELL TRAIT (SCT)

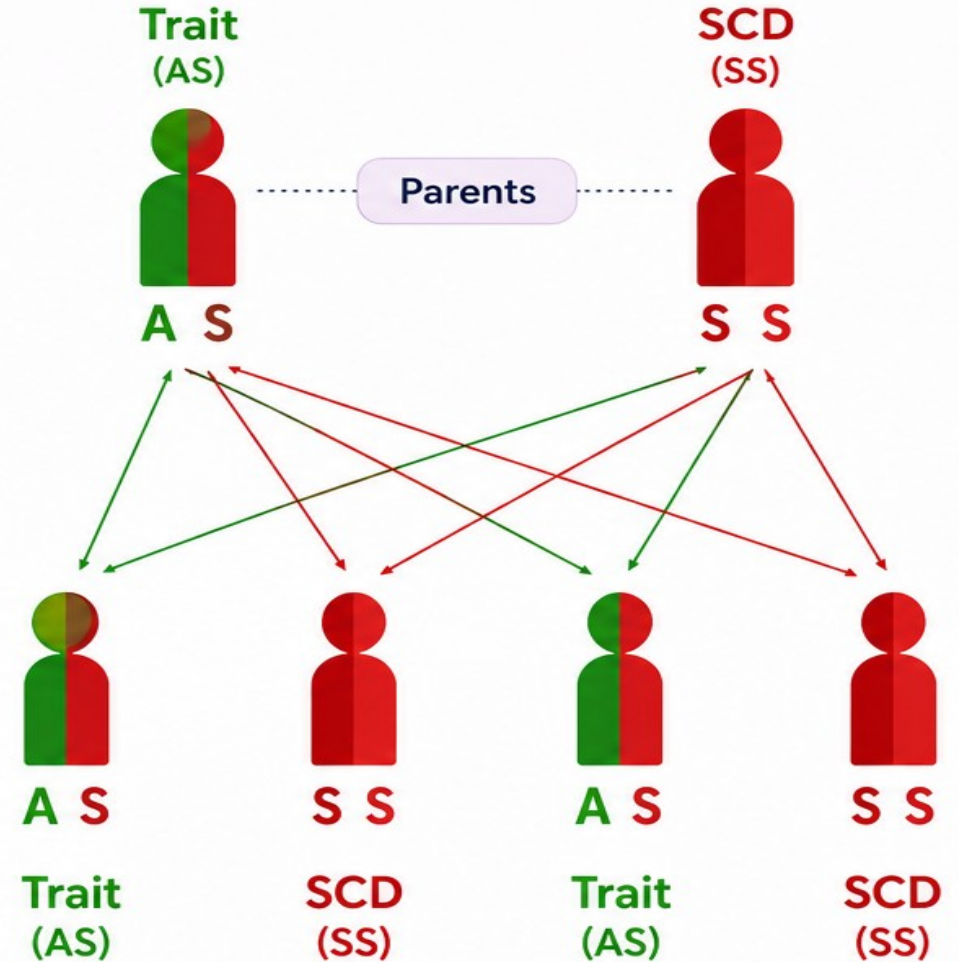
People who have SCT inherit one sickle cell gene (S) from one parent and one normal gene (A) from the other parent. People with SCT usually do not have any of the signs of the disease and live a normal life, but they can pass the trait to their children. Exposure to extreme stressful conditions, can lead to crisis which may require repeat check-ups.



HbS BETA THALASSEMIA

People who have this form of Sickle Cell disease inherit one sickle cell gene (S) from one parent and one gene for beta thalassemia from the other parent. Those with HbS beta thalassemia usually have a severe form of SCD.

INHERITANCE PATTERN



Effect of SCD on Mother and Baby

Mother

- Preterm labor and Spontaneous miscarriage
- Severe anemia due to hepatic sequestration crisis
- Antepartum and postpartum Hemorrhage
- Renal disease
- Hepatic dysfunction
- Infections
- Poor weight gain
- PIH & Pre-Eclampsia
- Acute chest syndrome
- Thromboembolic events
- Acute painful crisis
- Dysfunctional labor
- Maternal Death

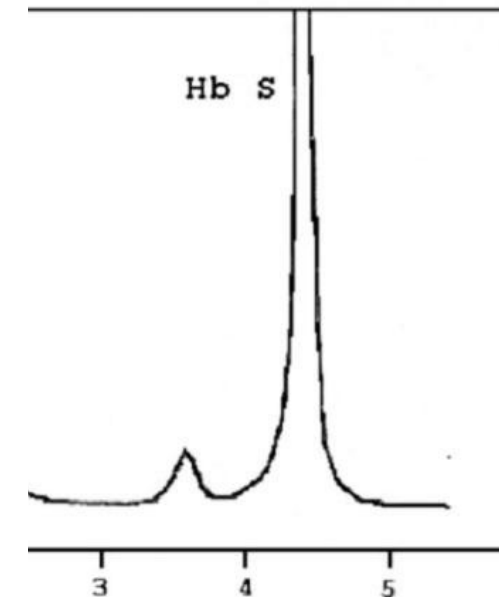
Baby

- Preterm labor and Spontaneous miscarriage
- Fetal growth restriction/ IUGR
- Abortions
- Prematurity
- Low birth weight
- Fetal loss
- Increased perinatal mortality
- Stillbirth

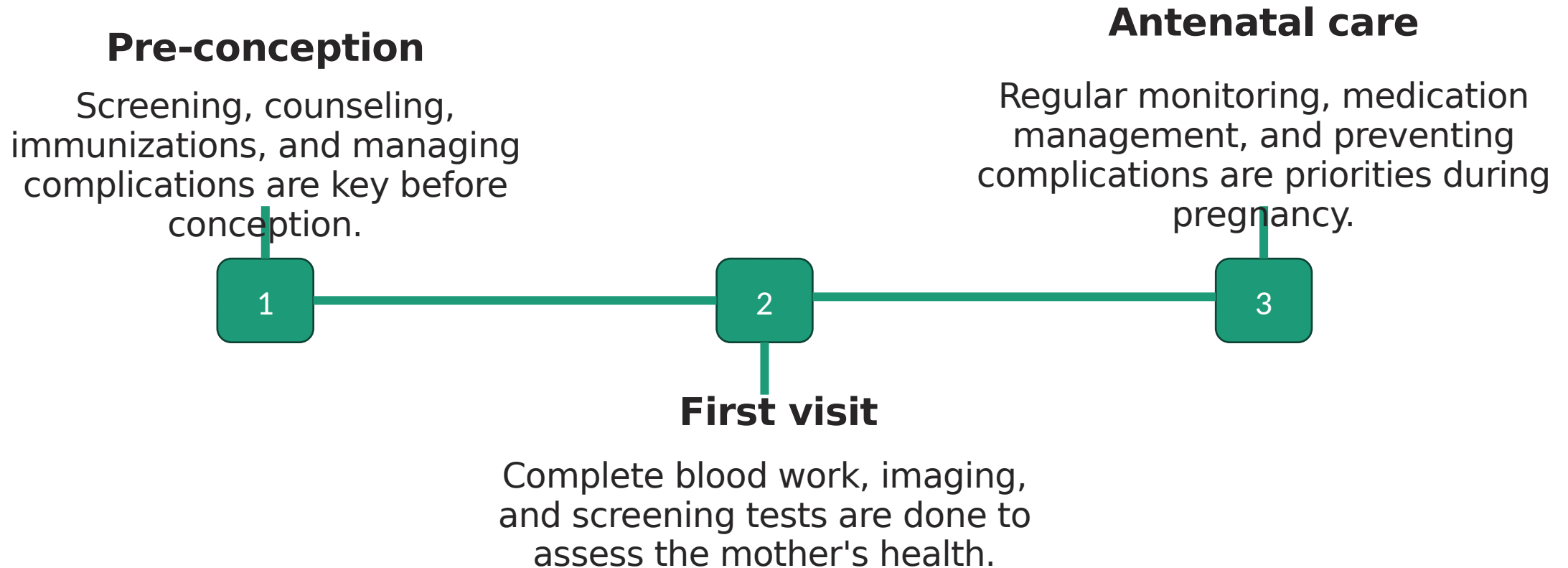
High Performance Liquid Chromatography

Principle: This method separates the hemoglobins into different fractions and identifies them based their retention time and shape of peak.

Criteria used for differentiating HPLC patterns.	
A > S	Sickle cell trait, sickle alpha-thalassemia
S, F and no A	Sickle cell anemia, Sickle-beta thalassemia
S > A and F	Sickle-beta thalassemia
A > C	HbC trait
C, F and no A	HbC disease, HbC-beta thalassemia
C > A	HbC-beta thalassemia



Prevention and Management of SCD



Prevention and Management of SCD

Pre-Conception period:

- ✓ Check the finding of sickle cell screening on the SC card .
- ✓ Provide premarital / pre-conceptual counselling and screening.
- ✓ Create awareness about the disease, planning for pregnancy and anticipating possible problems.
- ✓ Women with SCD if planning for pregnancy - should receive immunizations, Folic acid 5mg, (prophylactic penicillin if splenectomized).
- ✓ Routine assessments for complications of sickle cell disease and referred to higher centre if there are any complications of disease.

Medications to be stopped:

- ✓ Pre-conceptually Hydroxyurea should be stopped at least 3 months before conception.
- ✓ Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers should be stopped before conception.
- ✓ Once pregnant the iron chelation medicines will need to be stopped.

Assessment prior to conception/First visit post conception:

- ✓ Complete blood count with peripheral blood smear examination.
- ✓ Blood pressure and urinalysis should be performed.
- ✓ Renal and liver function tests should be performed annually to identify sickle nephropathy and/or deranged hepatic function.
- ✓ 2D Echo for pulmonary hypertension.
- ✓ Retinal Screening.
- ✓ Screening for iron overload before pregnancy by serum ferritin and T2* MRI imaging.
- ✓ If high iron overload is there, aggressive iron chelation (desferrioxamine) is advisable, before conception.
- ✓ Screening for red cell antibodies. Presence may indicate an increased risk of hemolytic disease of the newborn.
- ✓ Screening of partner/spouse for Sickle cell or Beta thalassemia trait.

Management of Sickle cell disease during Antenatal period

Medications:

- 5 mg Folic acid supplementation
 - Treat nausea/ vomiting of pregnancy early.
 - Continue antibiotics (Penicillin Prophylaxis).
 - Iron supplementation should be given if there is laboratory evidence of iron deficiency.
 - Low-dose aspirin 75 mg once daily from 12 weeks of gestation to reduce the risk of developing pre-eclampsia and thrombosis. Aspirin may be stopped at 36 weeks to reduce the risk of PPH.
 - Venous thromboembolism prophylaxis: Compression stocking use daily and prophylactic low-molecular-weight heparin 0.4 mg during antenatal hospital admissions.
 - NSAIDs should be prescribed only between 12 and 28 weeks of gestation owing to concerns regarding adverse effects on fetal development.
 - Vaccinations: influenza vaccine (If not received in previous year) Pneumococcal, HIB, Meningococcal and Hep B vaccines.
- ✓ Regular antenatal check-ups required.
 - ✓ Blood pressure and urinalysis should be performed at each consultation and midstream urine for culture performed monthly. The risk of pre-eclampsia is high.
 - ✓ Avoid precipitating factors of sickle cell crises such as exposure to extreme temperatures, dehydration and overexertion. Persistent vomiting can lead to dehydration and crisis. If this occurs, should receive IV fluids.

STOP: Hydroxyurea, Iron Chelators deferoxamine (DFO), ACE inhibitors/ARBs

Medications and Monitoring

Medications to Stop

Hydroxyurea, ACE inhibitors, and iron chelators should be stopped before and during pregnancy.

Medications to Continue

Folic acid, antibiotics, low-dose aspirin, and anticoagulants are important to manage during pregnancy.

Monitoring

Regular blood tests, ultrasounds, and screening for complications are crucial throughout pregnancy.

Management of Sickle cell disease during Antenatal period

Ultrasound examination:

- Offer a viability scan at 7–9 weeks of gestation.
- Offer routine first-trimester scan (11– 14 weeks of gestation) and a detailed anomaly scan at 20 weeks of gestation.
- May require serial fetal biometry scans (growth scans) every 4 weeks from 24 weeks of gestation.

Management of Sickle cell disease during Antenatal period

- Testing of spouse along with genetic counselling for all positive cases should be offered.
- If both couples have positive test, then refer for prenatal diagnostic services.

Prenatal Screening and Diagnosis



Chorionic Villus Sampling

This prenatal test can identify the sickle cell gene in the fetus during pregnancy.



Amniocentesis

This prenatal test can also be used to screen the fetus for sickle cell disease.



Genetic Counseling

Couples at risk should receive counseling to understand their options and make informed decisions.

Diagnostic Test of Sickle Cell

VHSND Level

Dry Blood Sample be done during 1st ANC at VHSND level. Report to be collected and shared with PW



Facility level

Gazella test can also be used for PW for sickle cell disease/trait.



PMSMA

Every PW has to be tested with Gazella Machine in the PMSMA

 **Mission Directorate**
National Health Mission, Odisha
Department of Health & Family Welfare,
Government of Odisha

Letter No.: OSH&FWS/ 5964 /2026
58788 Date: 09.06.26

From: Dr. Brundha D, IAS
Mission Director, NHM, Odisha

To: All CDM&PHO-Cum-DMOs, Odisha
All ADUPHOs (S MCOs), Odisha

Subj: Special initiative for sickle cell & thalassemia screening, treatment / management of pregnant women - Regarding

Madam/ Sir,

In inviting a reference to the subject cited above, it is requested to provide special focus on pregnancy cases for sickle cell & thalassemia screening, treatment / management. The detailed modalities of focus are as follows:

1. The Screening of pregnant women (PW) for sickle cell & thalassemia shall be done within one month of ANC Registration and preferably within 4 weeks of pregnancy.
2. The pregnant women (PW) shall be registered in National Sickle Cell Anaemia Elimination Mission (NSCAEM) portal using ABHA ID prior to the screening either at Field / Facility level.
3. The Screening shall be done using available screening kits at VHSND/uHSND level. In case of non-availability of the same, the self screening shall be done at the Facility level as early as possible after registration, using existing **Gazella machines** at the facility level.
4. Further, all the facilities conducting **PMSMA / e-PMSMA** sessions shall be equipped with **Gazella machines** for ensuring screening of pregnant women whose results are not available or have not been tested. In case of non-availability of Gazella machines, the same shall be procured within a month, utilizing XVFC (for primary health care facilities/ NSCAEM) funds.
5. The test results shall be reflected in the MCP card by Health Worker (Female). She shall obtain the result from the NSCAEM portal.
6. In case the result is not uploaded in the portal even after 15 days from the date of the screening, the same shall be intimated to the concerned BPM and the BPM in turn will e-mail the same to the Prog. Asst (Sickle Cell) / AM-ANCD for necessary action.
7. In case the **sample is rejected** for any reason, then repeat test shall be undertaken to ascertain the status.
8. All trait and disease cases (pregnant women) shall be marked as **HIGH RISK** and will be issued **RED card** at the point of detection. Treatment shall be initiated after first trimester at the nearest secondary care facility level.
9. In case of sickle cell trait/disease cases, screening of spouse shall be done and if both are found to be traity disease then **Chorionic villus sampling (CVS) testing** shall be done at nearest tertiary care facility (currently available at SCR MCH, Cutback & VMMSAR, Bhub). Further, appropriate steps shall be taken as per the guidance of the treating physician.
10. These cases shall be followed up as per HRP treatment protocol, and the status shall be recorded in HRP follow up Register at facility as well as Sub center level and PMSMA portal.

Modalities of screening, treatment and management

- The **Screening of pregnant women (PW) for sickle cell & thalassemia** shall be done within one month of ANC Registration and preferably within 4 weeks of pregnancy.
- The pregnant women (PW) shall be registered in **National Sickle Cell Anaemia Elimination Mission (NSCAEM) portal** using ABHA ID prior to the screening either at Field / facility level.
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Modalities of screening, treatment and management

- Further, all the facilities conducting **PMSMA/ e-PMSMA sessions shall be equipped with Gazelle machines** for ensuring screening of pregnant women whose results are not available or have not been tested. In case of non-availability of Gazelle machine, the same shall be **procured within a month**, utilizing XV-FC (for primary health care facilities)/ NIDAAN funds.
- The test results must be **reflected in the MCP card** by Health Worker (Female). She shall obtain the result from the **NSCAEM** portal.
- In case the result is not uploaded in the portal even after 15 days from the date of the screening, the same shall be intimated to the concerned BPM and the BPM in turn will e-mail the issue to the Prog. Asst (Sickle Cell)/ NCD consultant for necessary action.

Modalities of screening, treatment and management

- Birth preparedness and complication readiness plan must be discussed with each SCD positive woman and family. The pregnant woman should be guided to deliver at the nearby functional FRU following the referral protocol.
- Treatment shall be taken up as per NSCAEM guidelines. Proper counselling of such cases shall be done by RMNCHA+N counsellors at facility level and CHOs/ ANMs at community level.
- Also, a database shall be maintained in GOOGLE SHEET of all PW cases & their **screening, treatment / management status** till the specific NIC portal for the said purpose is ready.
- 104 call centre service can be availed by beneficiaries for more information on sickle cell & thalassemia disease.

Management of Sickle cell disease in 2nd & 3rd trimester

- Close monitoring of the mother's and baby's health.
- Careful ANC and supervision (every 15 days visit -12 to 28 weeks. Weekly after 28 weeks).
- Continue Folic acid – 5mg/day.
- Iron supplementation for only proven cases of iron deficiency anemia.
- The blood pressure and urine should be checked regularly.
- Continue Low dose Aspirin (75mg) to prevent pre-eclampsia between 12 to 36 weeks.
- Screening for UTI – Asymptomatic bacteriuria by Multi uristrix dip stick and midstream urine culture monthly.
- Hospitalizations – in cases of infections or unusual symptoms.

Acute Pain Crisis Management

1

Prompt Treatment

Provide pain relief, hydration, and oxygen within 30 minutes of reaching the hospital.

3

Multidisciplinary Care

A team of specialists should assess and manage the crisis to ensure the best outcome.

2

Diagnostic Tests

Complete blood count, cultures, and imaging can help identify the cause of the crisis.

4

Preventive Measures

Avoiding triggers like dehydration and extreme temperatures can help prevent crises.

Acute Pain Crisis Management

The commonest cause of admission with 27% and 50% of women having a painful crisis during pregnancy.

- Rest & advice to avoid excess heat/cold environment, exercise and dehydration.
- Adequate fluids/Hydration – oral /IV (60 ml/kg/day)
- If pain crisis, give analgesia within 30 minutes of reaching the hospital – NSAIDs if 12–28 weeks of gestation.
- Weak opioids such as Tramadol/codeine can be used for moderate pain, and stronger opiates such as morphine can be used for severe pain.
- If SpO₂ < 95 %, give oxygen.
- Refer to higher center.

Mode and timing of delivery

- SCD itself is not an indication for LSCS or contraindication for vaginal delivery.
- Mode and timing of delivery should depend on obstetric indications. Yet in most women with Hb SS disease, prefer to deliver soon after 37 weeks are reached, and surely before 39 weeks.
- Delivery to be planned at a higher institutions like DH/Medical college, which are able to manage both the complications of SCD and high-risk pregnancies.

Roles & Responsibilities of MO/OG

Medical Officer

- Screen pregnant women and spouse for sickle cell disease.
- Pre and post-test counselling.
- Referral of the identified cases to higher centers for diagnosis and treatment initiation.
- Refer to attend e-PMSMA clinic for further diagnostics and treatment.
- Follow-up of confirmed cases.